RETAIL SALES PERFORMANCE ANALYSIS

Using Microsoft Excel, Power Query and Power BI

Descriptive | Diagnostic | Predictive | Prescriptive

₹4.38M	1,500	24.8%	₹347.90K
Total sales	Orders	Return rate	Total discount

Dataset: Product Sales by Region | Data period: Jan 2023 to Jun 2025

Domain: Retail and Sales Analytics | 1,500 records, 5 regions, 4 stores, 7 products

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1 Project Overview and Objective

This project turns raw retail sales data into business insights using Microsoft Excel and Power BI. Excel and Power Query clean and prepare the data and give a first look with PivotTables. Power BI then builds the data model, the DAX calculations and an interactive dashboard on top of the prepared data.

Objective
To find sales trends, compare product and regional performance, and study discounts, returns and delivery, so that better business decisions can be made.

2 Data Source

Item	Details
Dataset name	Product Sales by Region
Source	excelx.com/practice-data/sales-retail/
Domain	Retail and Sales Analytics
Data period	1 January 2023 to 30 June 2025
Dataset size	1,500 records; 19 original attributes plus 1 calculated column (Delivery Time)

3 Problem Statement

1. Sales performance
Analyse sales across regions, products and customers and find the top and low performers.

2. Sales trend
Study sales over time and estimate the expected sales for 2025.

3. Discounts
Check whether discounts increase the quantity purchased and the sales value.

4. Returns and shipping
Analyse returns and shipping cost by product, region, store and salesperson.

5. Delivery time
Check whether slow delivery is linked to returns.

6. People and stores
Compare salesperson and store performance for rewards, training and improvement.

4 Attribute (Column) Details

One Sales Fact table and four Dimension tables

4.1 Sales Fact Table

Attribute	Data type	Description
Order ID	Text	Unique order reference
Order Date	Date	Date the order was placed
Delivery Date	Date	Date the order was delivered
Delivery Time	Whole number (calculated)	Days taken to deliver (Delivery Date minus Order Date)
Product ID	Text	Product code; links to Product Dim
Store ID	Text	Store code; links to Store Dim
Customer ID	Text	Customer code; links to Customer Dim
Quantity	Whole number	Units purchased
Unit Price	Decimal	Price per unit
Total Price	Decimal	Transaction value after discount
Discount	Decimal (%)	Discount applied (0%, 5%, 10% or 15%)
Returned	Numeric	1 = returned, 0 = not returned
Shipping Cost	Decimal	Shipping cost of the order
Salesperson	Text	Salesperson responsible for the sale
Region	Categorical	Sales region; links to Region Dim

4.2 Dimension Tables

Table	Attributes	Description
Customer Dim	Customer ID, Customer Name, Customer Type	Unique customers; type is Retail or Wholesale
Product Dim	Product ID, Product Name	Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet
Store Dim	Store ID, Store Location	Store A, B, C and D
Region Dim	Region, Region Manager	Central, East, North, South, West

5 Tools and Technologies

Microsoft Excel
Data cleaning, formatting, formulas, PivotTables and the first analysis.

Power Query
Data transformation, data type correction and data preparation.

Power BI
Data model, DAX measures, KPI cards, interactive visuals and dashboards.

6 Data Pre-processing (Excel and Power Query)

Making the raw data clean, consistent and ready for analysis

6.1 No duplicates

All 1,500 rows were checked across every column. No duplicates, and every Order ID is unique.

6.2 No missing values

No blank cells in 1,500 rows. Columns not needed (Promotion, Payment Method, Date) were removed.

6.3 Standardizing Data Formats

Field type	Columns	Standardization applied
Date fields	Order Date, Delivery Date	Consistent date format (dd-mm-yyyy)
Numerical fields	Quantity, Delivery Time, Returned	Whole number
Financial fields	Unit Price, Total Price, Shipping Cost	Currency format (₹)
Discount	Discount	Percentage format

6.4 Creating a Calculated Field (Excel)

A new column, **Delivery Time**, gives the number of days between the order date (B2) and the delivery date (C2).

```
=C2-B2
```

6.5 Filtering, Sorting and Structured Table

- Filters were applied by Region, Product ID, Store ID, Salesperson, Discount and Returned.
- Filters on Delivery Time and Shipping Cost were used to find slow deliveries and high shipping charges.
- Data was sorted by Order ID (ascending) and by Order Date to see sales activity over time.
- The dataset was converted into a structured Excel table, so headers stay available for filtering and sorting.

6.6 Fact and Dimension Worksheets

Data was organised into one Sales Fact table and four dimension tables (Customer, Product, Store, Region) to keep it neat and reduce repetition. **XLOOKUP** mapped Product IDs, Store IDs and Customer IDs. After the IDs were created, the text columns were removed from the fact table and kept only in the dimension tables.

```
=XLOOKUP(E2,$U$1:$U$8,$T$1:$T$8,"NOT FOUND")Product Name
=XLOOKUP(J2,$U$12:$U$16,$T$12:$T$16,"NOT FOUND")  Store Location
=XLOOKUP(K2,$U$19:$U$1390,$T$19:$T$1390,"NOT FOUND")    Customer Name
```

6.7 Conditional Formatting

Column	Formatting	Purpose
Total Price	Red-yellow-green colour scale	Show low to high sales value
Returned	Blue bold font	Flag returned orders
Shipping Cost	Red fill	Flag shipping charges above ₹40
Delivery Time	Red bold font	Flag slow deliveries of 6 days or more

6.8 Descriptive Statistics (Total Price)

Measure	Value	Measure	Value
Mean	2,919.99	Minimum	6.97
Median	2,174.72	Maximum	11,077.00
Standard deviation	2,522.79	Range	11,070.03
Skewness	0.97	Sum	4,379,992.43
Kurtosis	0.08	Count	1,500

The mean is higher than the median and the skewness is positive, so a few high-value orders pull the average up.

6.9 Pivot Table (Excel)

A. Descriptive analysis: sales by product and top 5 customers

Total sales by each product

Product Name	Count of Order ID	Sum of Quantity	Average of UnitPrice	Sum of Total Price
Chair	209	2122	₹320	₹6,22,589
Desk	207	2026	₹295	₹5,55,267
Laptop	226	2408	₹304	₹6,84,417
Monitor	212	2177	₹317	₹6,51,629
Phone	195	1970	₹271	₹4,97,163
Printer	211	2336	₹302	₹6,84,387
Tablet	240	2577	₹283	₹6,84,539
Grand Total	**1500**	**15616**	**₹299**	**₹43,79,992**

Top 5 customers by sales

Customer ID	Sum of Total Price
C1248	₹17,314
C1278	₹13,843
C150	₹16,087
C172	₹13,668
C510	₹12,744
Grand Total	**₹73,656**





B. Diagnostic analysis: discount impact, and delivery time with return rate by region

Impact of Discount on Order Quantity and Order Value

Discount	Count of Order ID	Average of Quantity	Average of Total Price
0%	397	10.2	3067
5%	368	10.6	3065
10%	379	10.4	2845
15%	356	10.5	2686
Grand Total	**1500**	**10.4**	**2920**

Average Delivery Time and Return Rate by Region

Region	Average of Delivery Time	Average of Returned
Central	5.9	23.3%
East	6.0	26.0%
North	6.4	22.7%
South	5.9	26.1%
West	6.0	26.1%
Grand Total	**6.0**	**24.8%**





C. Predictive analysis: regional and monthly sales trend, 2023 to 2025

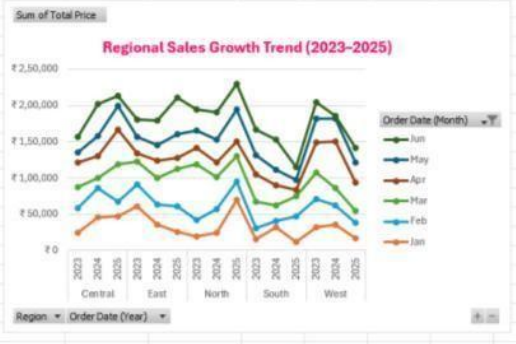
Regional Sales Growth Trend (2023-2025)

Sum of Total Price	Year						
Region	Jan	Feb	Mar	Apr	May	Jun	Grand Total
Central							
2023	₹ 24,975	₹ 34,140	₹ 28,906	₹ 33,237	₹ 13,626	₹ 22,121	₹ 1,57,005
2024	₹ 45,817	₹ 40,827	₹ 12,995	₹ 30,380	₹ 27,336	₹ 45,118	₹ 2,02,473
2025	₹ 46,770	₹ 20,395	₹ 51,932	₹ 47,280	₹ 33,336	₹ 13,850	₹ 2,13,563
East							
2023	₹ 60,935	₹ 29,992	₹ 31,240	₹ 11,914	₹ 22,282	₹ 23,995	₹ 1,80,359
2024	₹ 35,753	₹ 27,530	₹ 36,338	₹ 23,743	₹ 21,820	₹ 34,245	₹ 1,79,429
2025	₹ 25,564	₹ 35,831	₹ 51,711	₹ 14,490	₹ 32,335	₹ 50,953	₹ 2,10,884
North							
2023	₹ 19,544	₹ 22,656	₹ 76,111	₹ 23,770	₹ 23,058	₹ 28,956	₹ 1,94,095
2024	₹ 23,998	₹ 32,894	₹ 44,940	₹ 19,291	₹ 31,560	₹ 38,147	₹ 1,90,829
2025	₹ 69,354	₹ 25,501	₹ 35,810	₹ 20,173	₹ 43,404	₹ 35,199	₹ 2,29,440
South							
2023	₹ 15,726	₹ 14,688	₹ 36,456	₹ 38,541	₹ 25,634	₹ 36,066	₹ 1,67,111
2024	₹ 32,663	₹ 8,481	₹ 21,540	₹ 27,679	₹ 20,834	₹ 41,863	₹ 1,53,061
2025	₹ 12,349	₹ 35,157	₹ 27,883	₹ 7,721	₹ 14,884	₹ 16,980	₹ 1,14,972
West							
2023	₹ 31,632	₹ 39,881	₹ 35,835	₹ 41,669	₹ 32,666	₹ 23,087	₹ 2,04,771
2024	₹ 36,218	₹ 25,784	₹ 24,535	₹ 64,419	₹ 31,243	₹ 3,997	₹ 1,86,196
2025	₹ 17,062	₹ 21,301	₹ 15,858	₹ 39,012	₹ 28,390	₹ 19,685	₹ 1,41,308
Grand Total	**₹ 4,98,359**	**₹ 4,15,058**	**₹ 5,32,091**	**₹ 4,43,319**	**₹ 4,02,408**	**₹ 4,34,262**	**₹ 27,25,496**

Yearly and Monthly Sales Trend (2023-2025)

Sum of Total Price	Year			
Months	2023	2024	2025	Grand Total
Jan	₹ 1,52,812	₹ 1,74,449	₹ 1,71,098	₹ 4,98,359
Feb	₹ 1,41,357	₹ 1,35,516	₹ 1,38,185	₹ 4,15,058
Mar	₹ 2,08,549	₹ 1,40,348	₹ 1,83,194	₹ 5,32,091
Apr	₹ 1,49,131	₹ 1,65,512	₹ 1,28,675	₹ 4,43,319
May	₹ 1,17,266	₹ 1,32,793	₹ 1,52,350	₹ 4,02,408
Jun	₹ 1,34,225	₹ 1,63,370	₹ 1,36,667	₹ 4,34,262
Grand Total	**₹ 9,03,340**	**₹ 9,11,988**	**₹ 9,10,168**	**₹ 27,25,496**





D. Prescriptive analysis: store performance and sales by salesperson

Store Performance and Improvement Focus

Store Locatio	Sum of ShippingCc	Sum of Total Price	Average of Returned
Store A	₹ 10,285	₹ 10,88,513	24.7%
Store B	₹ 9,994	₹ 10,29,649	27.0%
Store C	₹ 9,910	₹ 10,56,305	22.5%
Store D	₹ 11,072	₹ 12,05,526	25.0%
Grand Total	**₹ 41,261**	**₹ 43,79,992**	**24.8%**



Total Sales by Salesperson

Salesperson	Sum of Total Price	Count of Order ID	Average of Total Price	Average of Returned
Bob	₹ 7,96,781	243	3279	26.3%
Alice	₹ 7,86,166	259	3035	24.7%
Frank	₹ 7,14,642	243	2941	22.6%
Carlos	₹ 7,07,167	270	2619	28.1%
Eva	₹ 6,98,670	236	2960	21.6%
Diana	₹ 6,76,568	249	2717	24.9%
Grand Total	**₹ 43,79,992**	**1500**	**2920**	**24.8%**



7 Data Modelling and DAX (Power BI)

From the cleaned Excel workbook to a star schema, columns and measures

7.1 From Excel to Power BI

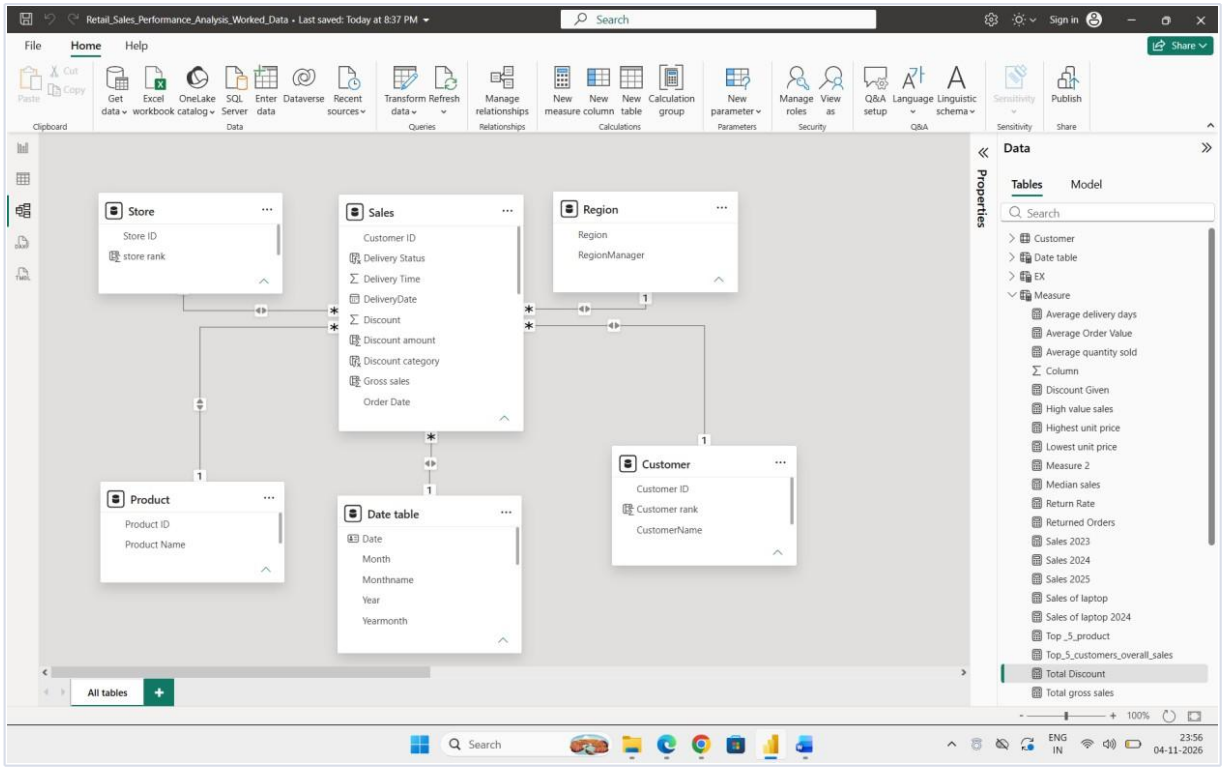
The cleaned workbook (Sales Fact and four dimension tables) was loaded with **Get data, Excel workbook**. Every PivotTable question has a matching column, measure or visual in Power BI.

Excel step	Power BI equivalent	Where it is used
Delivery Time column (=C2-B2)	Delivery Time, Delivery Status, Average delivery days	Delivery analysis
Fact and dimension sheets (XLOOKUP)	Star schema, one-to-many relationships	Data model
Total sales by product	EX table, Top_5_product, Product rank	Product-wise sales visuals
Top 5 customers	Customer rank, Top_5_customers_overall_sales	Descriptive page
Discount impact	Gross sales, Discount amount, Discount category, Total Discount	KPI cards, Diagnostic page
Returned by region	Return Status, Return Rate, Returned Orders	Return analysis visuals
Monthly and regional trend	Date table and time-intelligence measures	Predictive page
Store and salesperson pivots	Store rank, Total sales by Store Location	Prescriptive page

7.2 Data Model (Star Schema)

Sales is the central fact table, connected to the Product, Store, Customer and Region dimensions by one-to-many relationships. A calculated Date table connects through the order date, and a separate Measure table stores the report measures.

Table	Type	Key / main columns
Sales	Fact	Order ID, Order Date, Product ID, Store ID, Customer ID, Region, Quantity, UnitPrice, Discount, Total Price, ShippingCost, Returned, Delivery Time
Product	Dimension	Product ID, Product Name
Store	Dimension	Store ID, StoreLocation
Customer	Dimension	Customer ID, CustomerName, CustomerType
Region	Dimension	Region, RegionManager
Date table	Dimension (calculated)	Date, Year, Month, Monthname, Yearmonth



7.3 What Each Measure Tells the Business

Every measure answer one simple business question. The color shows which analysis type uses it.

Measure	Question it answers	Result	Used for
Total sales	How much money did we earn?	₹4.38M	**Descriptive**
Total orders	How many orders did customers place?	1,500	**Descriptive**
Average Order Value	How much does one order bring?	₹2.92K	**Prescriptive**
Unique customers	How many different people bought?	1,371	**Descriptive**
Total Discount	How much did we give away as discount?	₹347.90K	**Diagnostic**
Return Rate	How many orders come back?	24.8% (372 orders)	**Diagnostic**
Average delivery days	How long does a customer wait?	6.04 days	**Diagnostic**
Total shipping cost	How much do we spend on delivery?	₹41.3K	**Prescriptive**
Sales 2023 / 2024 / 2025	How did each year do?	₹1.70M / ₹1.77M / ₹910.17K (Jan to Jun)	**Predictive**
MoM growth	Did this month beat last month?	% change	**Predictive**
Sales YTD	How much have we sold so far this year?	₹1,771,955 at end of 2024	**Predictive**
Same period last year	Are we doing better than last year?	Year-over-year view	**Predictive**
Rolling 3 months	What is the recent trend?	3-month running total	**Predictive**

7.4 Formulas Used (DAX)

Date table (calculated table): one row per day from 1 Jan 2023 to 30 Jun 2025 (912 rows). Connect Date table[Date] to Sales[Order Date] and sort Monthname by Month.

```
Date table =
ADDCOLUMNS(
 CALENDAR(DATE(2023,1,1), DATE(2025,6,30))
 ,"Year", YEAR([Date]),
 "Month", MONTH([Date]),
 "Monthname", FORMAT([Date], "mmm
 "),"Yearmonth", FORMAT([Date], "yyyy-
 mmm")
)
```

Column	DAX formula	What it gives
Gross sales	`Sales[Quantity] * Sales[UnitPrice]`	Sales value before discount (total ₹4.73M)
Discount amount	`Sales[Gross sales] - Sales[Total Price]`	Money given as discount (total ₹347.90K)
Discount category	`IF(Sales[Discount] >= 0.10, "High discount", "Low discount")`	High discount when 10% or more
Return Status	`IF(Sales[Returned] = 1, "Returned", "Not Returned")`	372 Returned, 1,128 Not Returned
Delivery Status	`SWITCH(TRUE(), Sales[Delivery Time] <= 3, "Fast", Sales[Delivery Time] <= 5, "Normal", "Slow")`	Fast (3 days or less), Normal (4 to 5), Slow (6 or more)
Sales Category	`SWITCH(TRUE(), Sales[Total Price] < 2000, "Low", Sales[Total Price] < 5000, "Moderate", Sales[Total Price] < 10000, "High", "Premium")`	Four order-value bands
Shipping category	`IF(Sales[ShippingCost] > 40, "High shipping", "Standard shipping")`	Flags shipping above ₹40
Order since	`DATEDIFF(Sales[Order Date], TODAY(), YEAR)`	Years since the order date

7.5 Ranking columns

```
Product rank  = RANKX(ALL('Product'), CALCULATE(SUM(Sales[Total Price])), , DESC, Dense)
store rank    = RANKX(ALL('Store'),   CALCULATE(SUM(Sales[Total Price])), , DESC, Dense)
Customer rank = RANKX(ALL(Customer),  CALCULATE(SUM(Sales[Total Price])), , DESC, Dense)
```

Rank	Product	Total sales	Store	Total sales
1	Tablet	₹684,539	Store D	₹1,205,526
2	Laptop	₹684,417	Store A	₹1,088,513
3	Printer	₹684,387	Store C	₹1,056,305
4	Monitor	₹651,629	Store B	₹1,029,649
5	Chair	₹622,589		
6	Desk	₹555,267		
7	Phone	₹497,163		

Customer rank ranks all 1,371 customers by their total sales.

7.6 Calculated tables

```
EX = SUMMARIZE(Sales, 'Product'[Product Name], "Total Sales", SUM(Sales[Total Price]))

Top_5_product =
TOPN(5,
 SUMMARIZE(Sales, 'Product'[Product Name], "Total Sales", SUM(Sales[Total Price])),
 [Total Sales], DESC)

Product data = FILTER(Sales, RELATED('Product'[Product Name]) = "Laptop")
```

EX gives sales for 7 products (matches the Excel pivot). **Top_5_product** gives Tablet, Laptop, Printer, Monitor and Chair. **Product data** keeps only the 226 Laptop rows.

7.7 Base report measures (Measure table)

Measure	Formula	Result
Total sales	SUM(Sales[Total Price])	₹4.38M
Total gross sales	SUM(Sales[Gross sales])	₹4.73M
Total orders	DISTINCTCOUNT(Sales[Order ID])	1,500
Total quantity	SUM(Sales[Quantity])	15.62K
Total Discount	SUM(Sales[Discount amount])	₹347.90K
Return Rate	AVERAGE(Sales[Returned])	24.8%
Returned Orders	SUM(Sales[Returned])	372
Average Order Value	DIVIDE([Total sales], [Total orders])	₹2.92K
Average delivery days	AVERAGE(Sales[Delivery Time])	6.04
Average quantity sold	AVERAGE(Sales[Quantity])	10.4
Median sales	MEDIAN(Sales[Total Price])	₹2.17K
Highest unit price	MAX(Sales[UnitPrice])	₹599.72
Lowest unit price	MIN(Sales[UnitPrice])	₹5.52
Unique customers	DISTINCTCOUNT(Sales[Customer ID])	1,371
Total shipping cost	SUM(Sales[ShippingCost])	₹41.3K
Current time / Today date	NOW() / TODAY()	Date and time cards

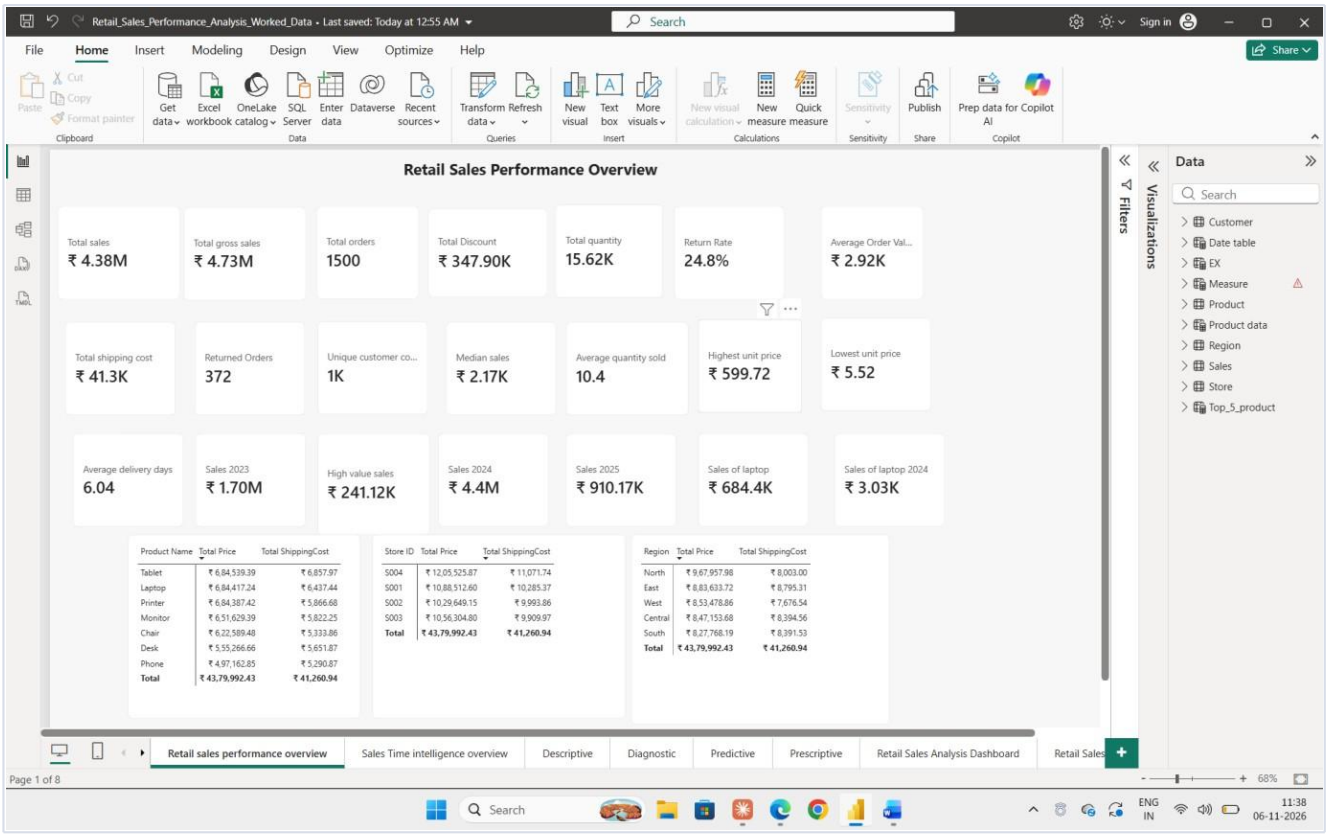
7.8 Time-intelligence measures

Measure	DAX formula	What it gives
Previous month sales	CALCULATE(SUM(Sales[Total Price]), PREVIOUSMONTH('Date table'[Date]))	Sales of the month before
MoM Growth	DIVIDE(SUM(Sales[Total Price]) - [Previous month sales], [Previous month sales])	% change against the previous month
Sales mtd	TOTALMTD(SUM(Sales[Total Price]), 'Date table'[Date])	Month-to-date sales
Sales ytd	TOTALYTD(SUM(Sales[Total Price]), 'Date table'[Date])	Year-to-date sales (₹1,771,955 at end of 2024)
Same period last year	CALCULATE(SUM(Sales[Total Price]), SAMEPERIODLASTYEAR('Date table'[Date]))	Same dates one year earlier
Date in period	CALCULATE(SUM(Sales[Total Price]), DATESINPERIOD('Date table'[Date], LASTDATE('Date table'[Date]), -3, MONTH))	Rolling 3-month total
Sales 2023 / 2024 / 2025	CALCULATE([Total sales], 'Date table'[Year] = 2023) (2024, 2025 the same way)	₹1.70M / ₹1.77M / ₹910.17K
Sales of laptop	CALCULATE([Total sales], 'Product'[Product Name] = "Laptop")	₹684.4K

8 Power BI Report Pages and Interactivity

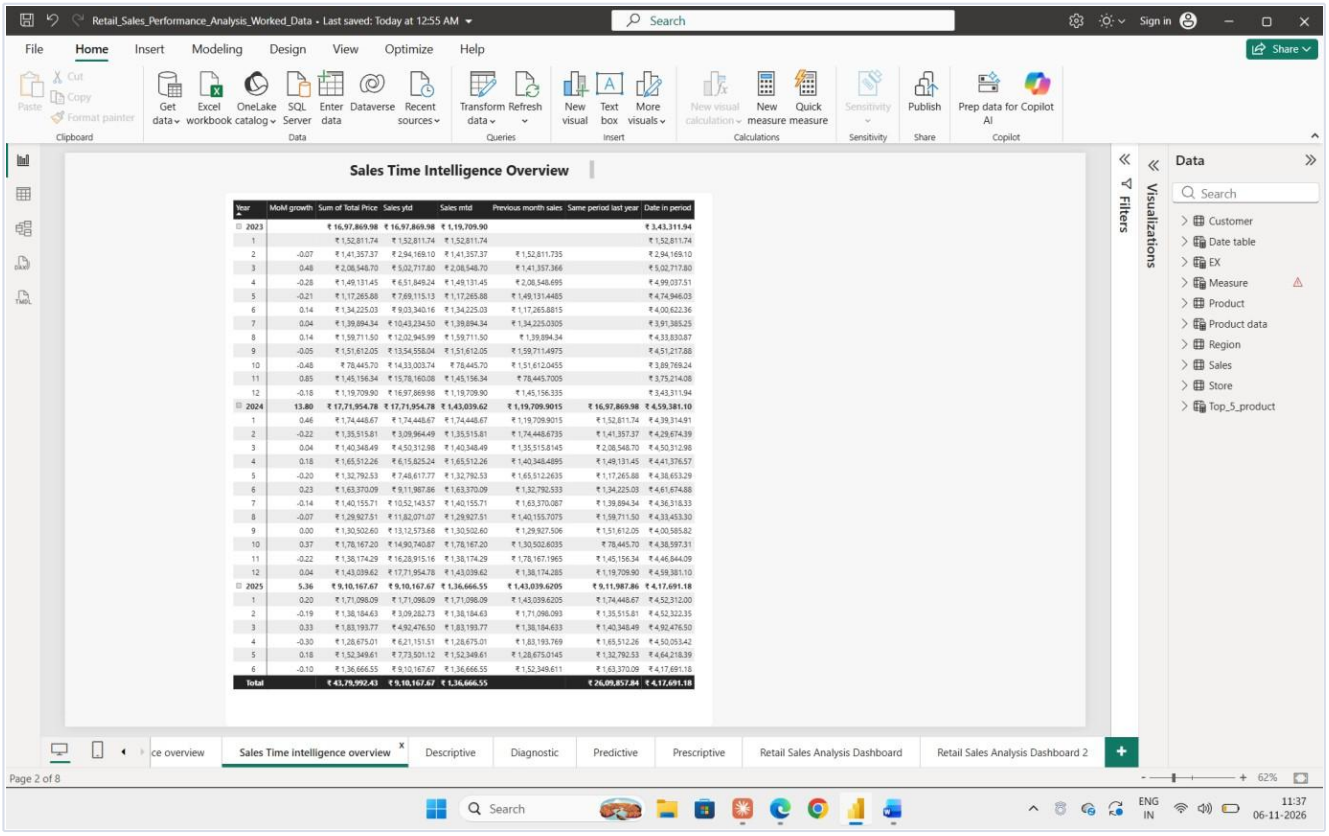
8.1 : Retail Sales Performance Overview

- KPI cards: Total sales, Gross sales, Orders, Discount, Quantity, Return Rate, Average Order Value, Returned Orders, Shipping cost, Average delivery days.
- Summary tables: total sales and shipping cost by product, by store and by region.



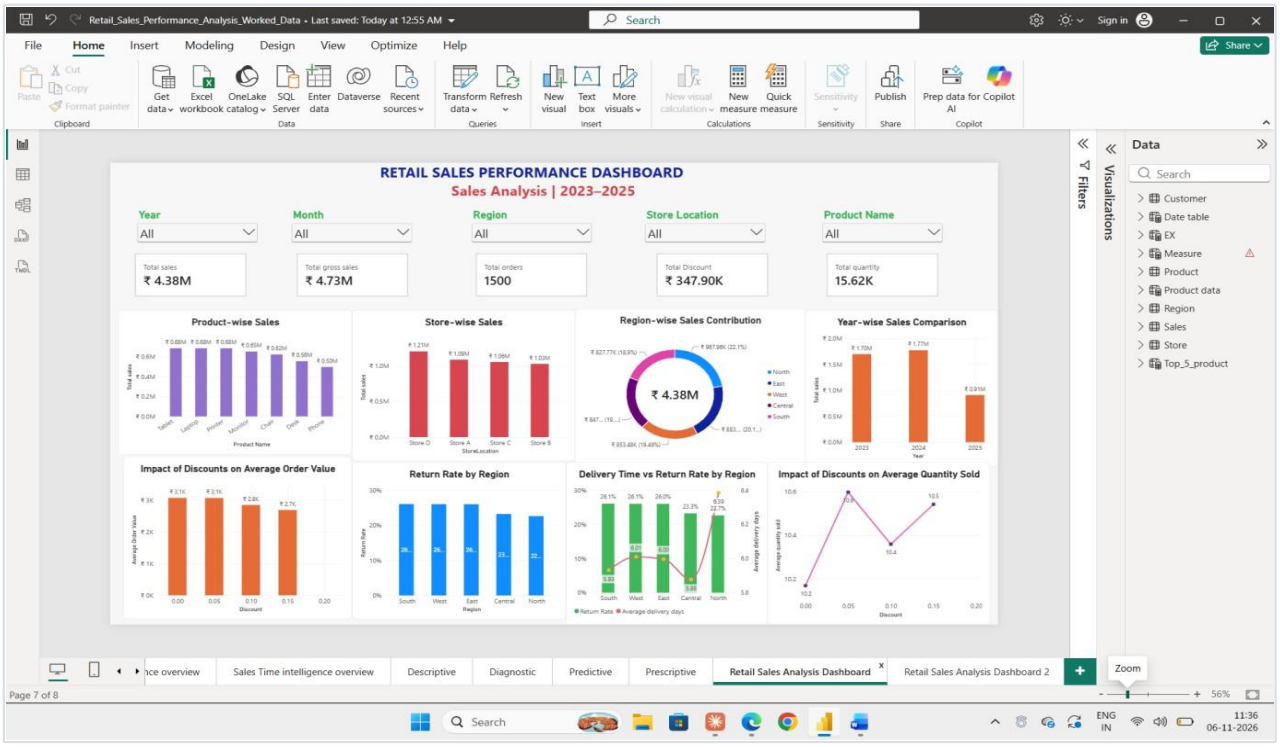
8.2 : Sales Time Intelligence Overview

- A matrix by year and month with Total Price, MoM growth, Sales YTD, Sales MTD, Previous month sales, Same period last year and the rolling 3-month total.



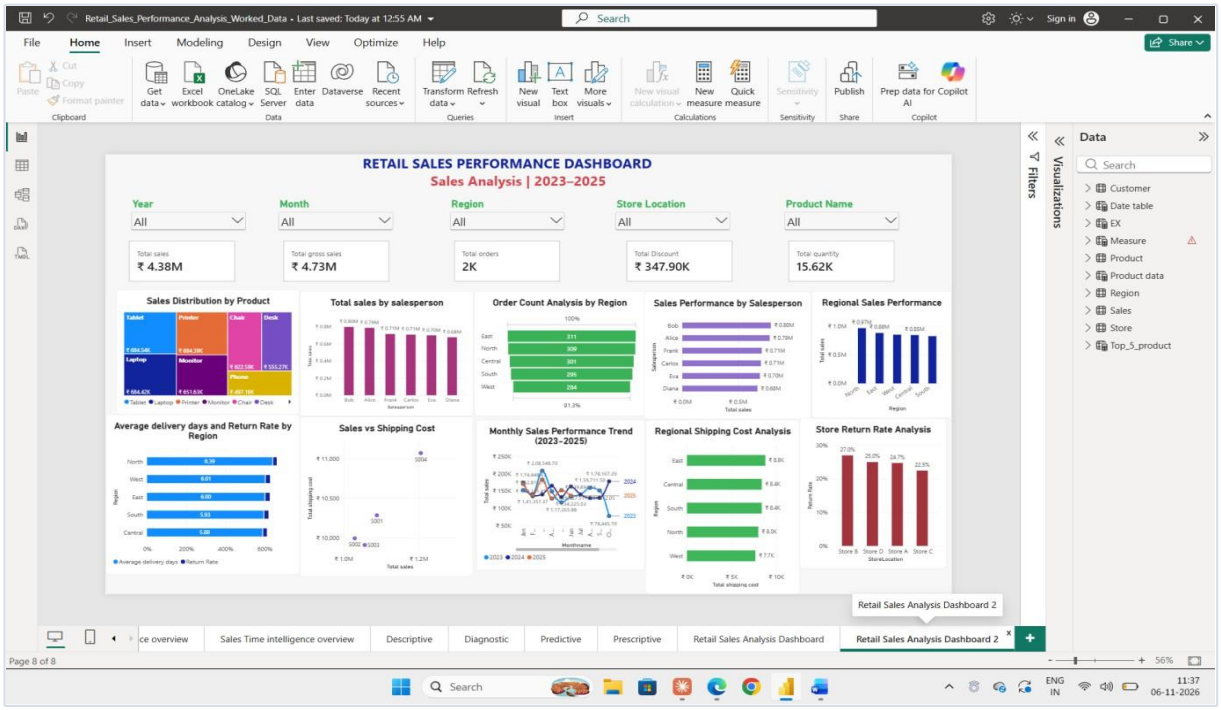
8.3 : Retail Sales Analysis Dashboard 1

Visual	Type	Purpose
Year, Month, Region, Store Location, Product Name	Dropdown slicers	Filter every card and chart on the page
Total sales, Gross sales, Orders, Discount, Quantity	Cards	Key figures at a glance
Product-wise, Store-wise, Year-wise Sales	Bar and column charts	Compare products, stores and years
Region-wise Sales Contribution	Donut chart	Share of each region
Discount impact, Return Rate by Region, Delivery Time vs Return Rate	Column and line charts	Check the effect of discounts, returns and delivery



8.4 : Retail Sales Analysis Dashboard 2

- Sales Distribution by Product (treemap), Total sales by salesperson and Order Count Analysis by Region.
- Sales Performance by Salesperson, Regional Sales Performance, Average delivery days and Return Rate by Region.
- Sales vs Shipping Cost, Monthly Sales Performance Trend, Regional Shipping Cost and Store Return Rate Analysis.



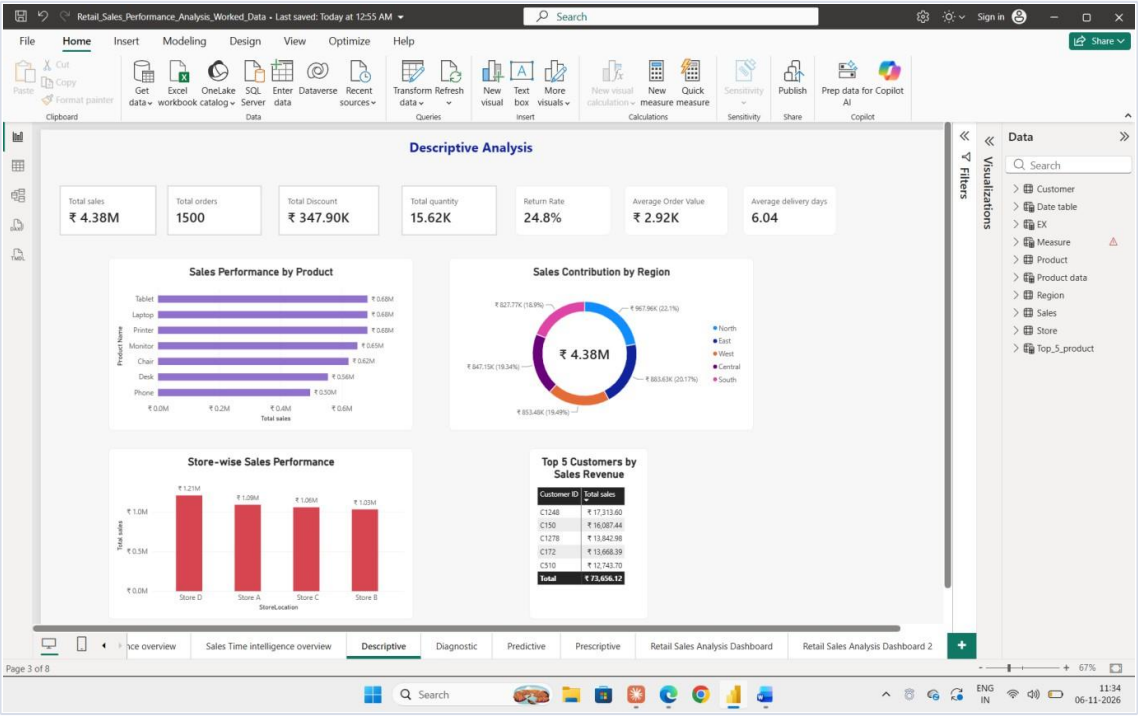
8.5 Interactivity

- Slicers filter all visuals on the page, and visuals cross-filter each other.
- The Year slicer has 2023, 2024 and 2025 (to June 2025), and months run from Jan to Dec.
- Titles, data labels and legends keep every visual clear, with one colour theme across pages.

9 Insights: The Four Types of Analysis

9.1 What happened? (Descriptive Analysis)

₹4.38M	1,500	1,371	24.8%
Total sales	Orders	Unique customers	Return rate

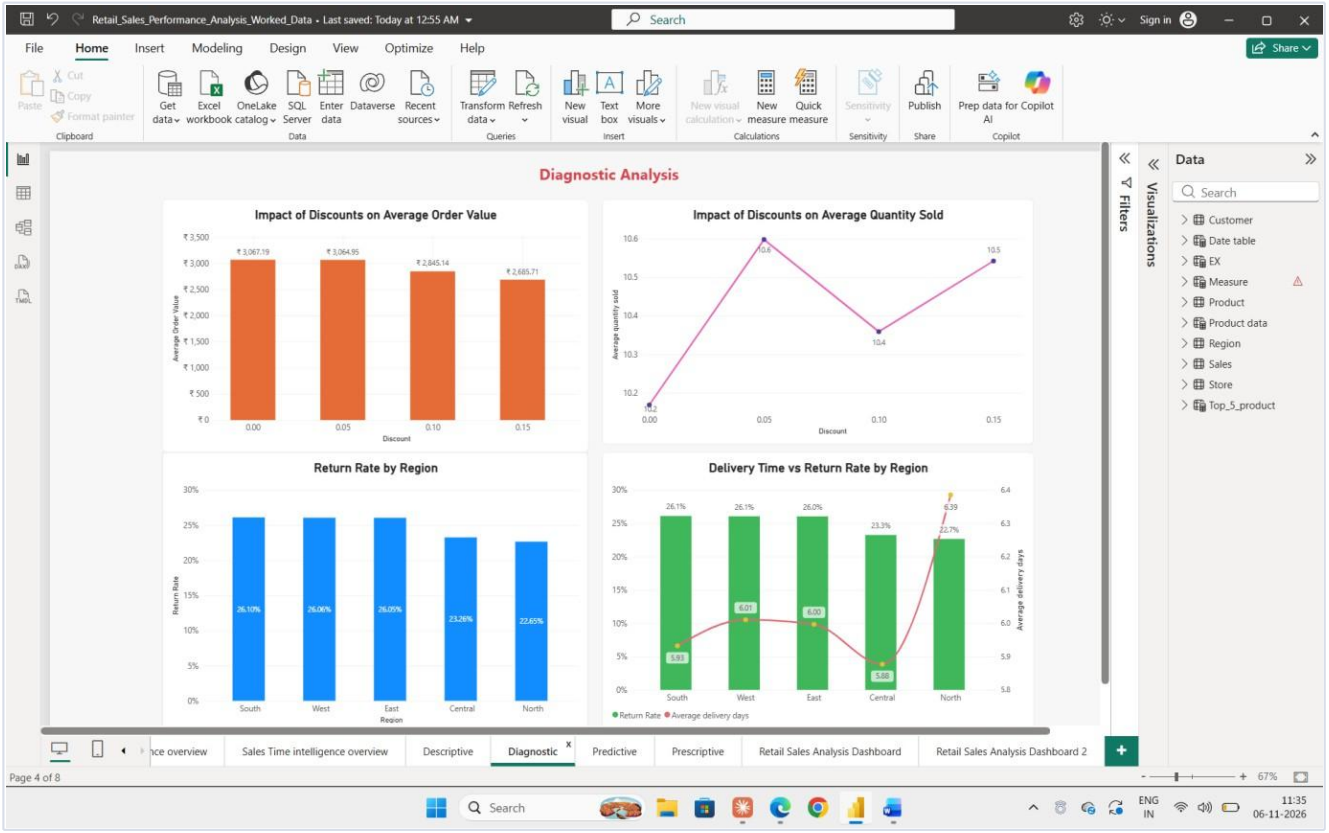


- The business made **₹4.38M** from **1,500 orders** and 1,371 customers.
- Tablet, Laptop and Printer are our stars at about ₹684K each. Phone sells the least (₹497K).
- North is our best region (₹968K) and South is our weakest (₹828K). Store D leads all stores with ₹1.2M.
- Top customers: C1248 (₹17,314), C150 (₹16,087) and C1278 (₹13,843). Wholesale (₹2.21M) and Retail (₹2.17M) are almost equal.
- But 372 of the 1,500 orders (24.8%) were returned.

> **Key insight**
> Tablet, Laptop and Printer bring about ₹684K each, while Phone brings only ₹497K. North (₹968K) leads and South (₹828K) is last. But 372 of the 1,500 orders (24.8%) were returned.

9.2 Why did it happen? (Diagnostic Analysis)

₹347.90K	10.4	6.04 days	24.8%
Total discount given	Avg units per order	Avg delivery time	Return rate



Whether the discount is 0%, 5%, 10% or 15%, customers buy about 10 units per order. Only the average order value goes down, from ₹3,067 to ₹2,686.

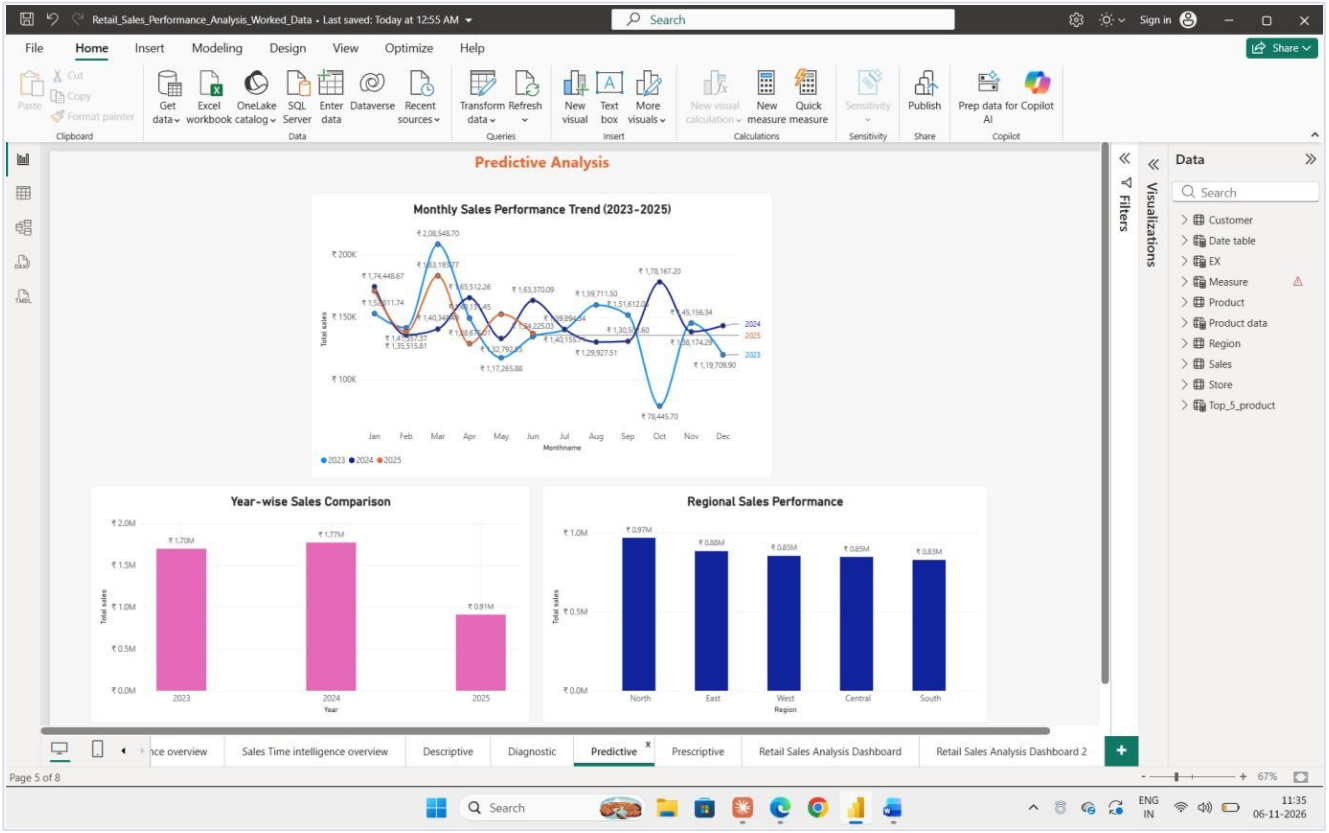
North has the slowest delivery (6.4 days) but the lowest return rate (22.7%). Central is the fastest (5.9 days), yet its return rate is higher (23.3%).

Key insight

Discounts do not make customers buy more, they only reduce our order value (₹3,067 to ₹2,686). Slow delivery is also not the reason for returns, because North delivers the slowest but has the fewest returns (22.7%).

9.3 What will happen? (Predictive Analysis)

₹1.70M	₹1.77M	₹910K	₹1.77M
Sales 2023	Sales 2024 (+4%)	Jan to Jun 2025	Expected 2025



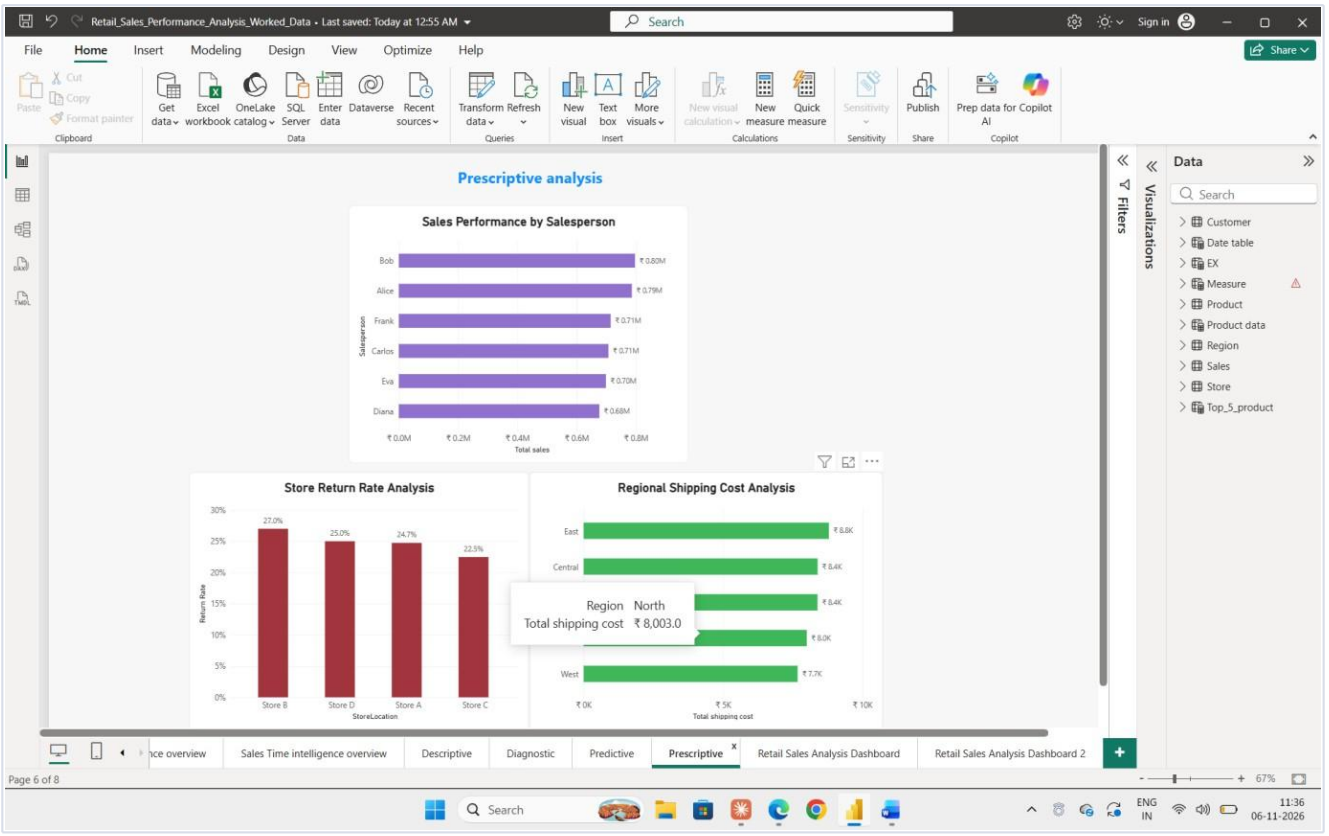
- Sales grew about 4% from 2023 (₹1.70M) to 2024 (₹1.77M).
- January to June 2025 (₹910K) is almost the same as January to June 2024 (₹912K), so 2025 should end near **₹1.77M** (range ₹1.71M to ₹1.82M).
- Under the surface, North is up 20% and East 18% (Central +5%), while South is down 25% and West 24%.

Key insight

2025 should end near ₹1.77M, flat against 2024. But North (+20%) and East (+18%) are growing while South (-25%) and West (-24%) are falling, so the gains and losses cancel out.

9.4 What should we do? (Prescriptive Analysis)

₹797K	28.1%	₹28.45	27.0%
Top seller: Bob	Carlos return rate	South shipping per order	Store B return rate



Reward
Bob (₹797K) and Alice (₹786K), our top sellers.

Train
Diana (lowest sales, ₹677K) and Carlos (most orders, but lowest order value and most returns).

Rethink discounts
They do not increase the quantity bought.

Fix South
It sells less and has the highest shipping cost per order (₹28.45).

Check Store B
It has the highest return rate (27.0%).

Key insight
Reward Bob (₹797K) and Alice (₹786K), and train Diana (lowest sales, ₹677K) and Carlos (highest return rate, 28.1%). Review the discount strategy, South (highest shipping cost, ₹28.45 per order) and Store B (highest return rate, 27.0%).

10 Conclusion

Conclusion

- The project combined Excel and Power BI to analyse retail sales, from data preparation to interactive reporting.
- Excel and Power Query cleaned the data, built the fact and dimension tables and gave a first analysis with PivotTables.
- Power BI added the data model, DAX columns, calculated tables and measures, and a dashboard with slicers, KPI cards and charts.
- The analysis covered sales trends, regional performance, products, discounts, delivery and returns.
- The business should review its discount strategy, improve sales in South and West, train lower-performing salespeople and investigate returns at Store B.
- Overall, the project shows how data analytics supports better business decisions.